VTI Academy

VISUAL STUDIO CODE DEVELOPMENT WITH AI INTEGRATION

Time: April 2026

Author: Huy Thanh Le

AGENDA

1. Say Hi !!!
2. Introduction Courses
3. VS Code Overview & Setup
4. Introduction to Github Copilot
5. Basic Application with Copilot Assistants
6. Recap and Homework Session_1

1. SAY HI !!!

SAY HI !!! - MY INFOR

My turn is first, …

+ My name is Huy (Henry)
+ Technical Lead AI/ML @IMT Solution
+ MsC Computer Science @Back Khoa
+ Domain Fields: Advanced Data Analytics and Scientist, ML/AI Research, AI Agentic Applications, NLP vs LLM Research
+ Contacts: +84 902167727 (Zalo)

And, what about you?...

2. Introduction Course

Introduction Syllabus Course - Main Infor

Purpose Course: This course is designed to provide participants with **practical, hands-on experience** in using AI tools—especially **GitHub Copilot**—to improve coding productivity and software development workflows, ***but we can move beyond if we can***

Durations: **3 days (12 hours total)**, with each session **4 hours** focusing on a combination of **theory and real-world practice**.

Throughout the course, participants will:

- Learn the fundamentals of **VS Code and AI-assisted development**
- Apply **prompt engineering techniques** for coding tasks
- Build a **real application with AI support**
- Develop a **RAG-based chatbot** for internal document search
- And **other AI Applications**

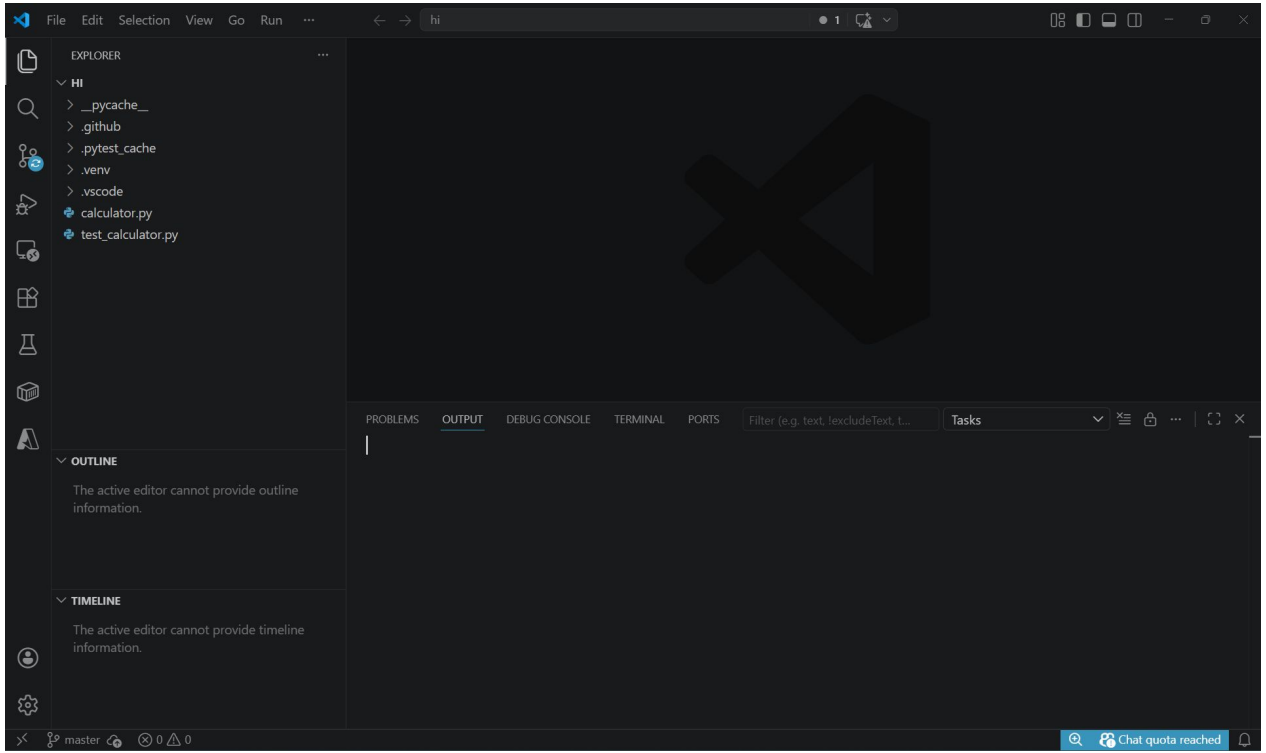
Link Detail: Click Here

3. VS Code Overview & Set up

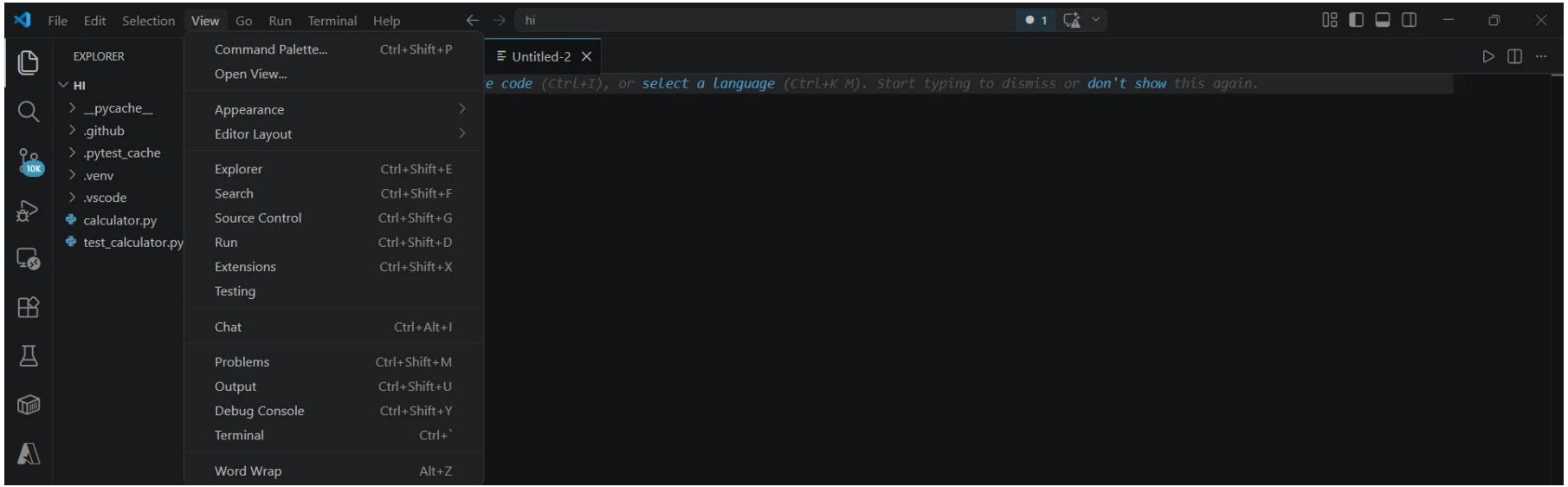
VS Code Overview - What is it?

VTI Academy

VS Code is code editors, with several key features belows



VS Code Overview - Command Palette



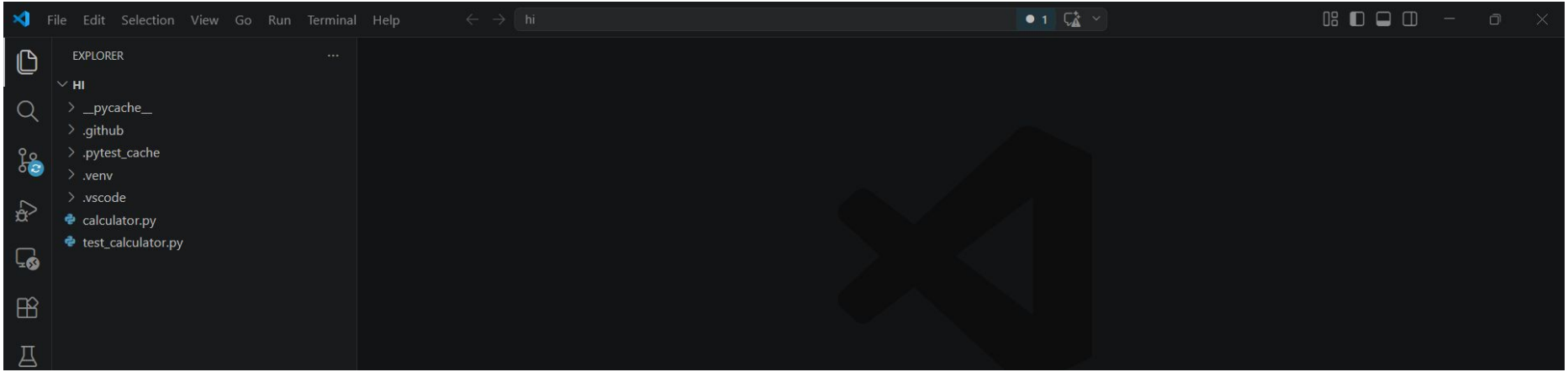
Command Palette: A quick access tool to search and run commands in VS Code.

Use it for: Access tools install in VS Code (Copilot, Chat, Git, etc.)

Exercise: Open Command Palette → Search for a command `Git: Clone`

Short Cut in VS Code: Ctrl + Shift + P

VS Code Overview - Explorer

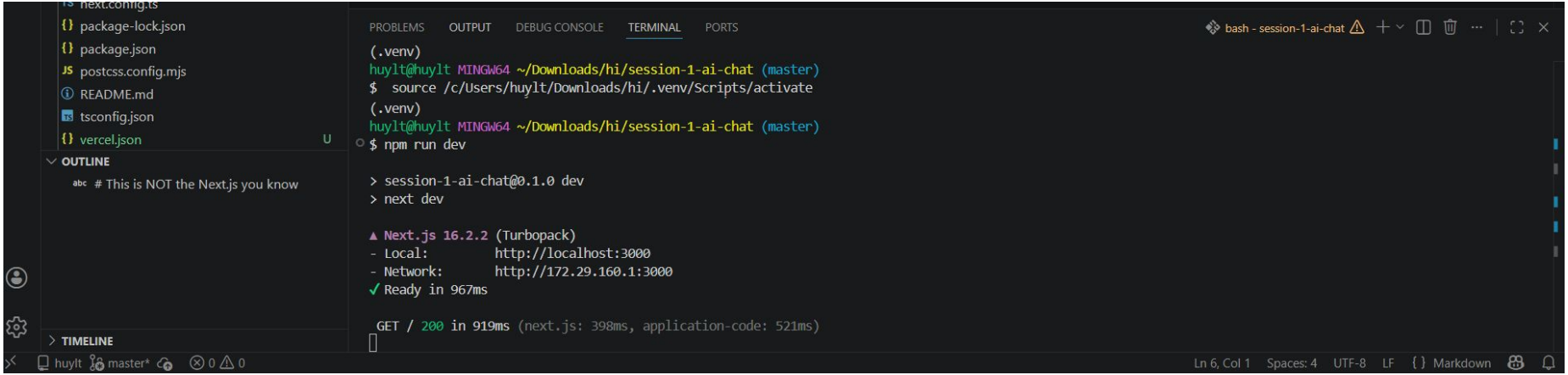


VS Explorer: Shows your project files and folders, allowing you to open, create, rename, and organize files easily.

Use it for: Managing project structure, navigating files, and organizing code.

Exercise: Create a folder → add a file (main.py) → rename it → delete it.

VS Code Overview - Terminal

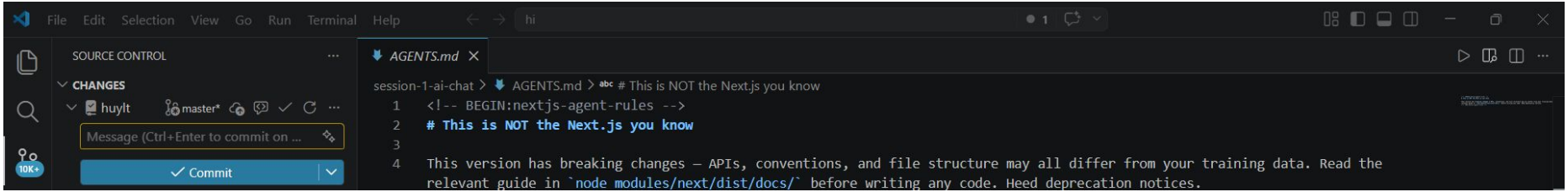


Terminal: Runs commands directly inside VS Code for building, running, and managing your project.

Use it for: Running apps, installing packages, managing environments, and debugging.

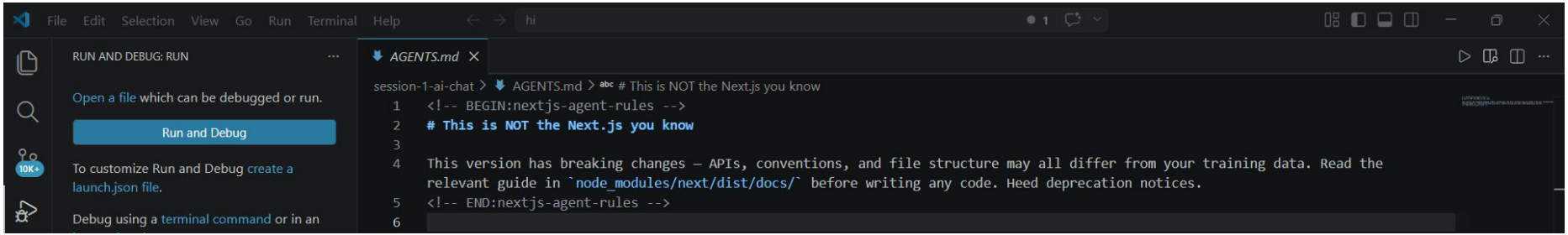
Exercise: Open terminal → create virtual env → activate it → pip install flask npm install packages axios

VS Code Overview - Source Code x Debugger



Source Code: Lets you write, edit, and manage your code with syntax highlighting and IntelliSense.

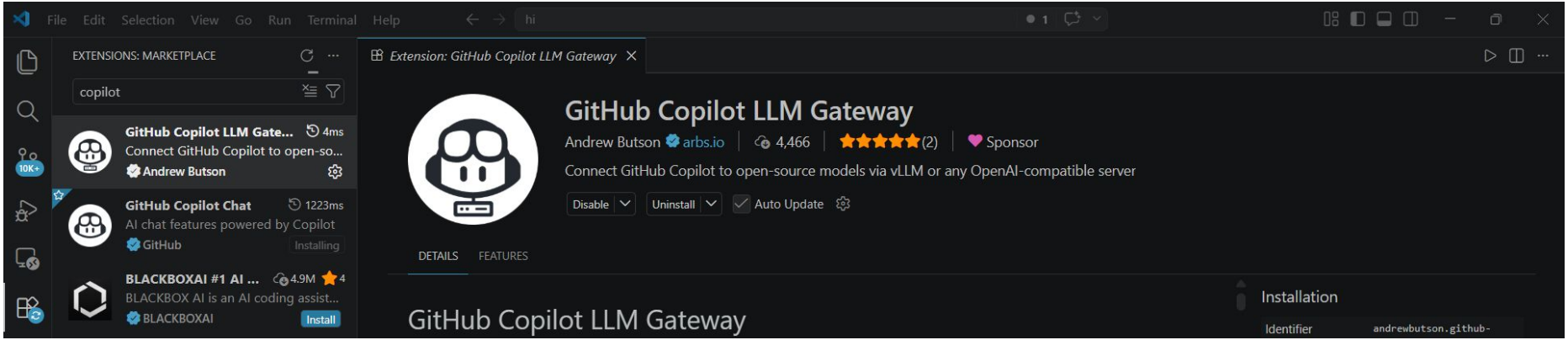
Use it for: Writing code, navigating files, and making changes efficiently.



Debugger: Helps you run code step-by-step to find and fix errors.

Use it for: Setting breakpoints, inspecting variables, and debugging logic.

VS Code Overview - Extensions and Languges Server



1. Extensions: Add new features to VS Code (e.g., Copilot, Prettier, Python).

Use it for: Improving productivity, code formatting, AI assistance, and tooling.

2. Language Server: Provides smart features like auto-complete, error checking, and code navigation.

Use it for: IntelliSense, linting, go-to definition, and real-time feedback.

Exercise: Install an extension (e.g. Python) → enable it → write code → use auto-complete → fix errors with suggestions.

4. Github Copilot

Github Copilot - What is it?

- Cloud-based generative
- AI tool Converts natural language prompts into coding suggestions
- Supports multiple programming languages
- Trained on billions of lines of code Acts like an AI pair programmer
- Works across various IDEs

Link: [New chat · GitHub Copilot]

Link: [GitHub Copilot in VS Code]

Github Copilot - It Helps What? and Not

Github Copilot - Planing Bill #1

Take flight with GitHub Copilot

For individuals | For businesses

Free

A fast way to get started with GitHub Copilot.

$0 USD

Get started

Open in VS Code

What's included:

- 50 agent mode or chat requests per month
- 2,000 completions per month
- Access to Haiku 4.5, GPT-5 mini, and more
- Copilot CLI

Pro Most popular

Accelerate workflows with GitHub Copilot.

$10 USD

per user / month

Try for 30 days free

Everything in Free and:

- Copilot cloud agent
- Copilot code review
- Claude and Codex on GitHub and VS Code
- 300 premium requests to use latest models, with the option to buy more[1]
- **Unlimited** agent mode and chats with GPT-5 mini[2]

Pro+

Scale with agents and more models.

$39 USD

per user / month

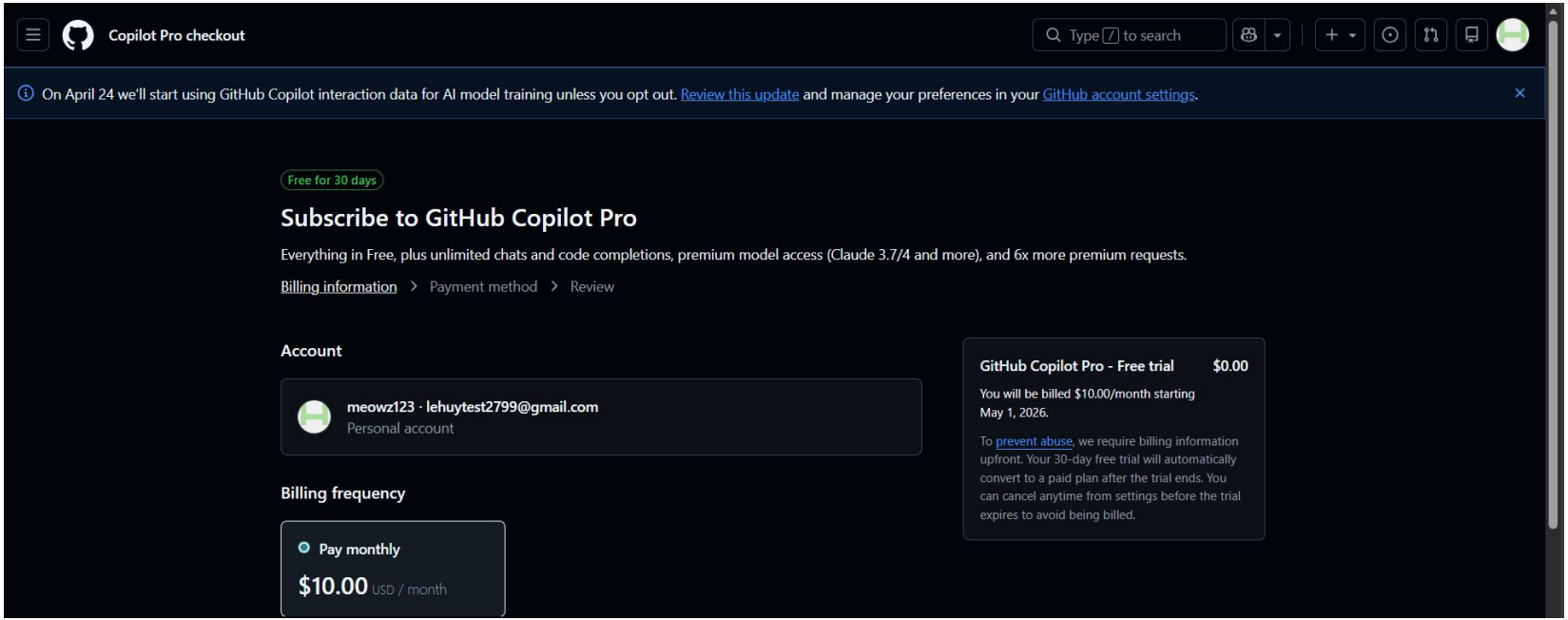
Get started

Everything in Pro and:

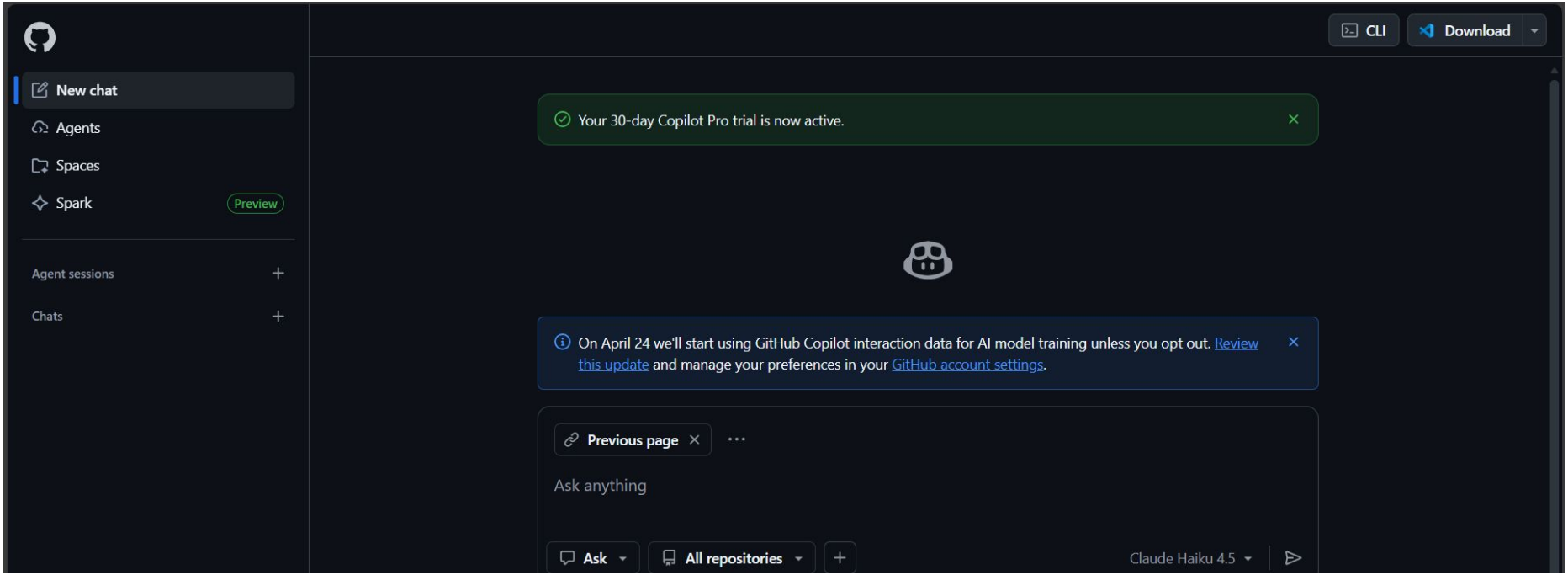
- Access to all models, including Claude Opus 4.6 and more
- 5× as many premium requests as Pro to use the latest models, with the option to buy more [1]
- Access to GitHub Spark

Link: [GitHub Copilot · Plans & pricing]

Github Copilot - Planing Bill #2



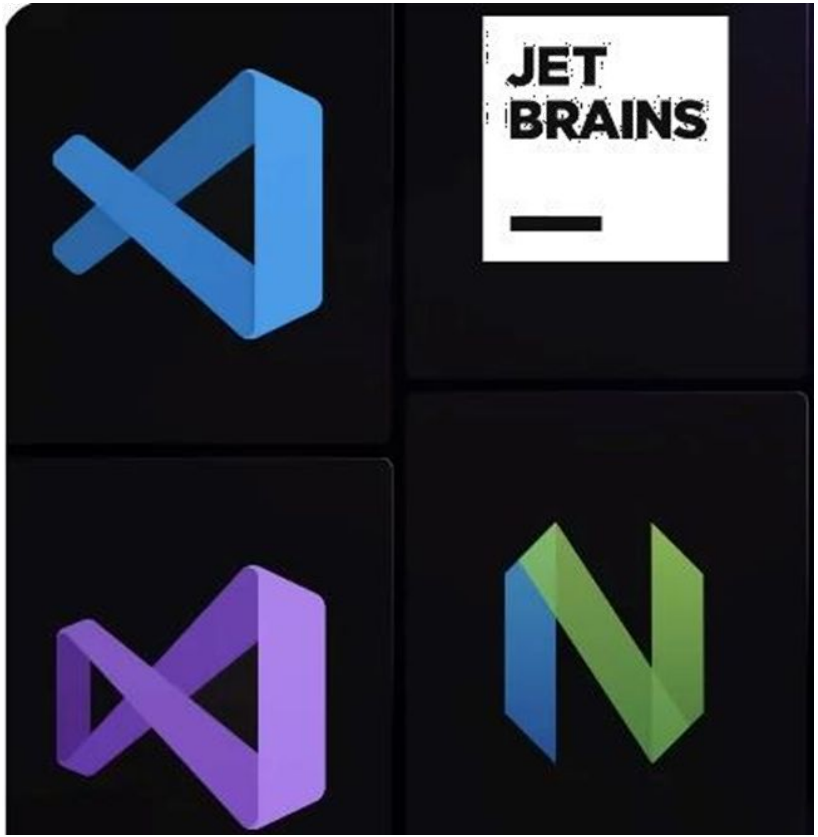
Github Copilot - Planing Bill #3



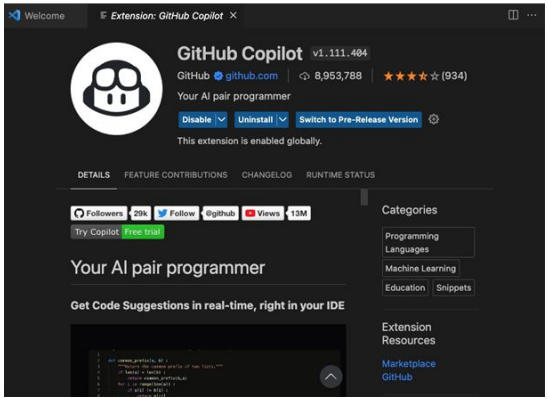
Github Copilot x IDE VS Code

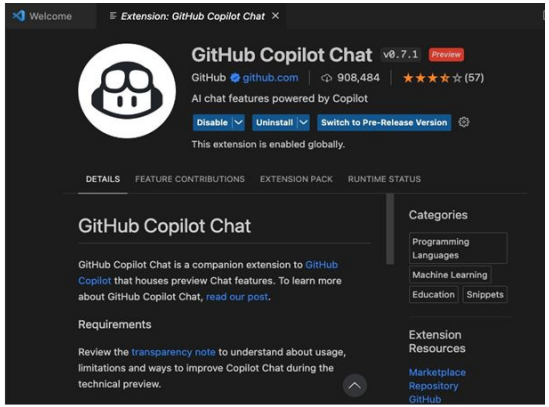
GitHub Copilot in **VS Code Professional 22** / IDEs

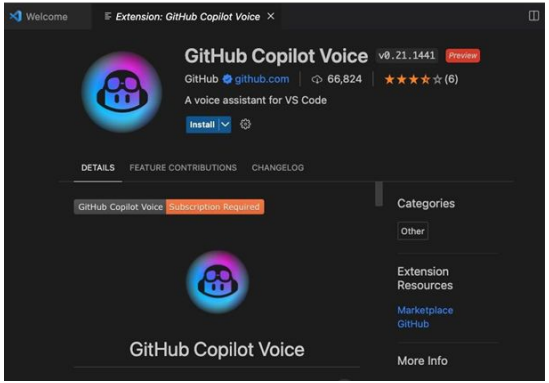
- Works directly inside VS Code and other IDEs (JetBrains, Neovim)
- Provides real-time code suggestions
- Supports multiple programming languages
- Helps generate functions, comments, and boilerplate code
- Improves productivity and reduces manual coding effort
- Easy integration with popular development environments

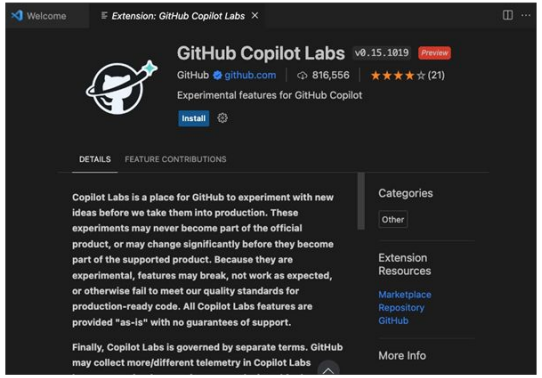


Github Copilot x VS Code Extension #1

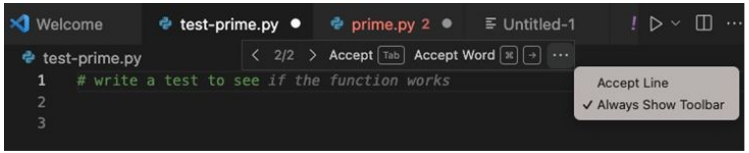


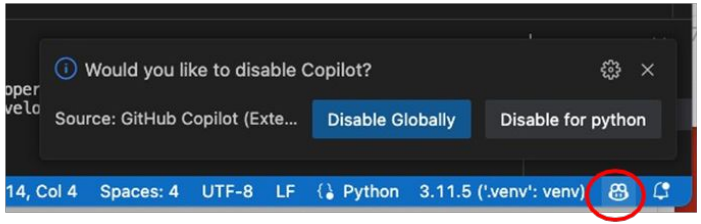






Github Copilot x VS Code Extension #2





Keyboard shortcuts for macOS

Action	Shortcut
Accept an inline suggestion	`Tab`
Dismiss an inline suggestion	`Esc`
Show next inline suggestion	`Option (⌥) or Alt + ]`
Show previous inline suggestion	`Option (⌥) or Alt + [`
Trigger inline suggestion	`Option (⌥) + \`
Open GitHub Copilot (additional suggestions in separate pane)	`Option (⌥) or Alt + Return`

Keyboard shortcuts for Windows

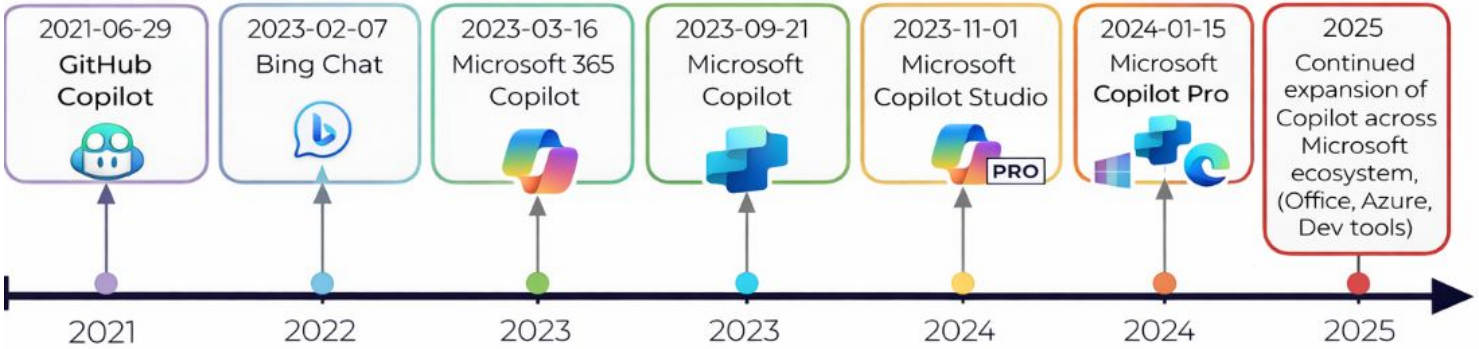
Action	Shortcut
Accept an inline suggestion	`Tab`
Dismiss an inline suggestion	`Esc`
Show next inline suggestion	`Alt + ]`
Show previous inline suggestion	`Alt + [`
Trigger inline suggestion	`Alt + \`
Open GitHub Copilot (additional suggestions in separate pane)	`Alt + Enter`

Keyboard shortcuts for Linux

Action	Shortcut
Accept an inline suggestion	`Tab`
Dismiss an inline suggestion	`Esc`
Show next inline suggestion	`Alt + ]`
Show previous inline suggestion	`Alt + [`
Trigger inline suggestion	`Alt + \`
Open GitHub Copilot (additional suggestions in separate pane)	`Alt + Enter`

Github Copilot History

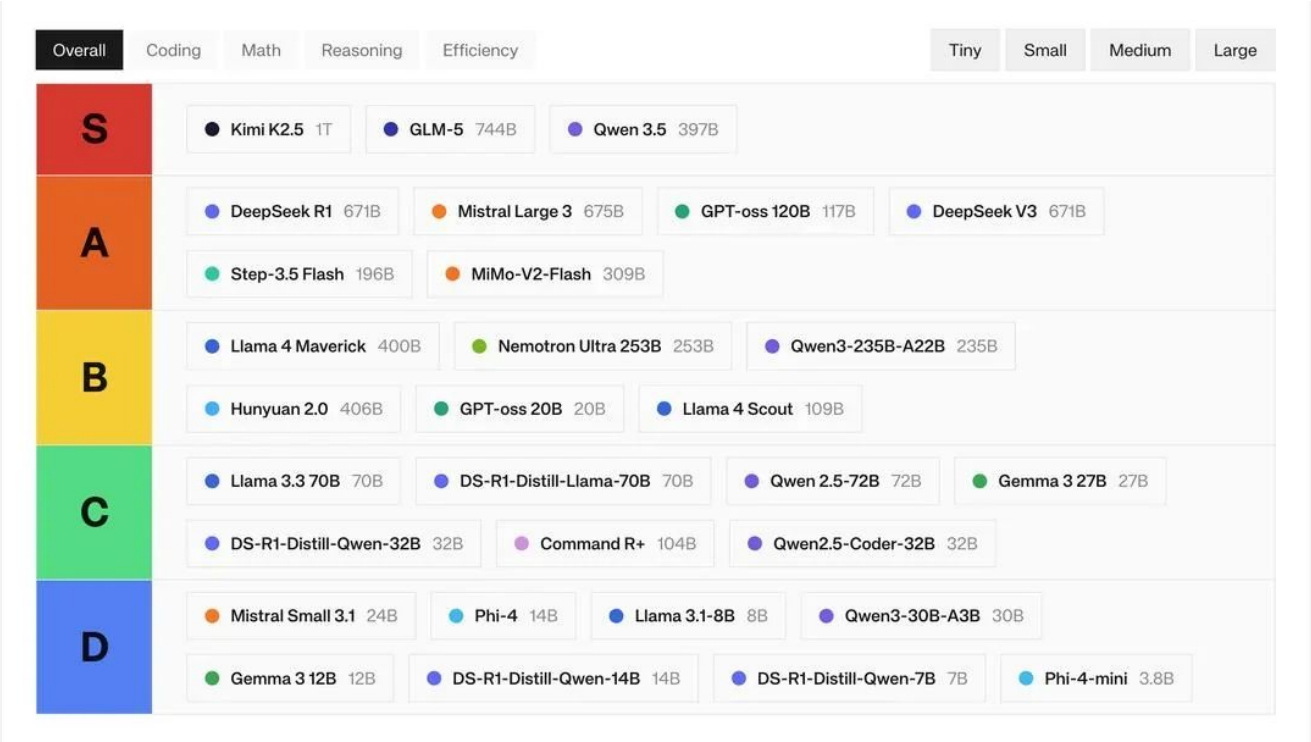
- 2021-06-29: GitHub Copilot
- 2023-02-07: Bing Chat
- 2023-03-16: Microsoft 365 Copilot
- 2023-09-21: Microsoft Copilot
- 2023-11-01: Microsoft Copilot Studio
- 2024-01-15: Microsoft Copilot Pro
- 2024-05 Copilot+ PCs AI-powered Windows devices
- 2024-06 GitHub Copilot Workspace (preview)
- 2024-11 Copilot enhancements across Windows&Edge



Github Copilot - AI Agentic Architecture

Github Copilot - AI Agent Brains

- **LLM** (Large Language Model) = AI that understands & generates text/code
- **Parameters** = size of AI brain (more = smarter but slower)
- **AI Powe**r = ability to write, debug, and explain code
- **How it's built:** Pretrained on internet + code
  Fine-tuned for coding



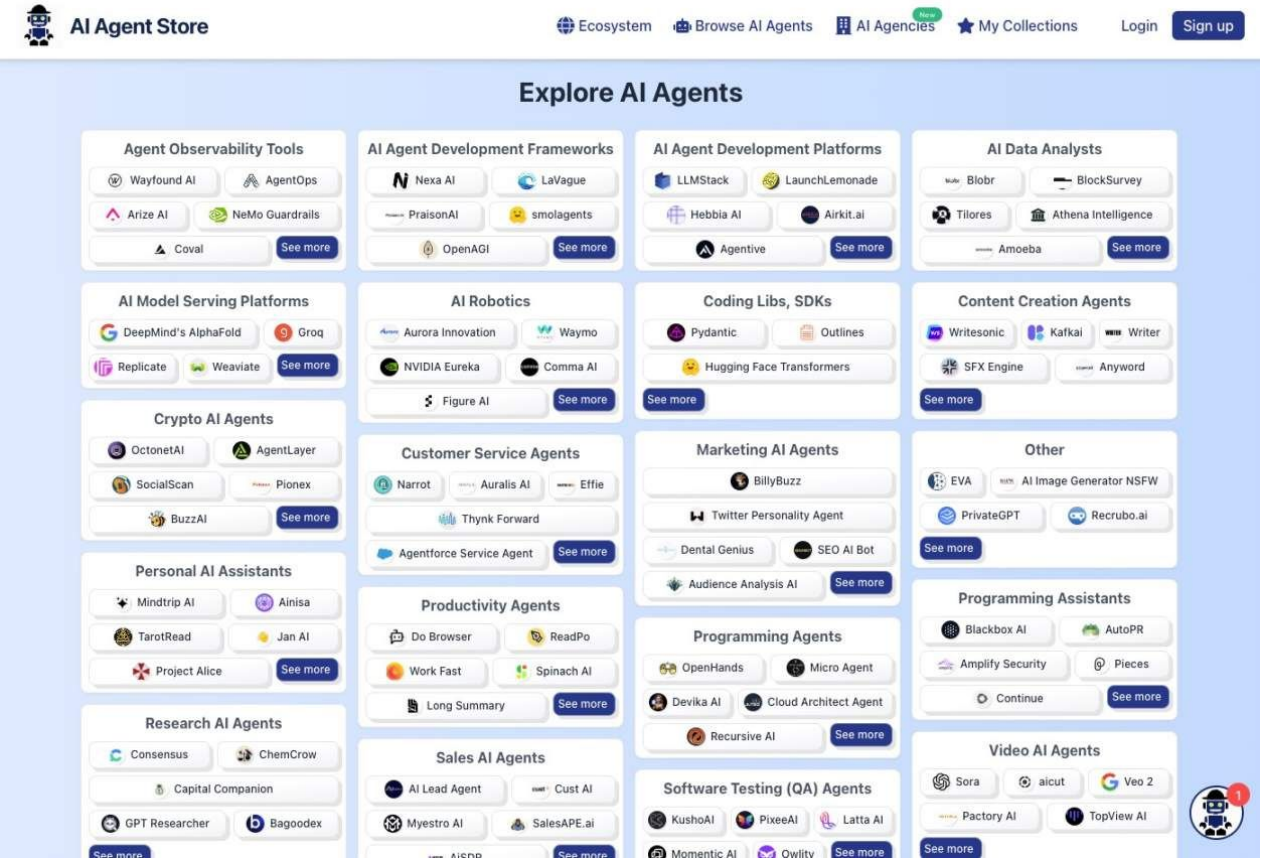
Link: [SWE-Bench Verified Leaderboard]

Github Copilot - AI Agent Hands

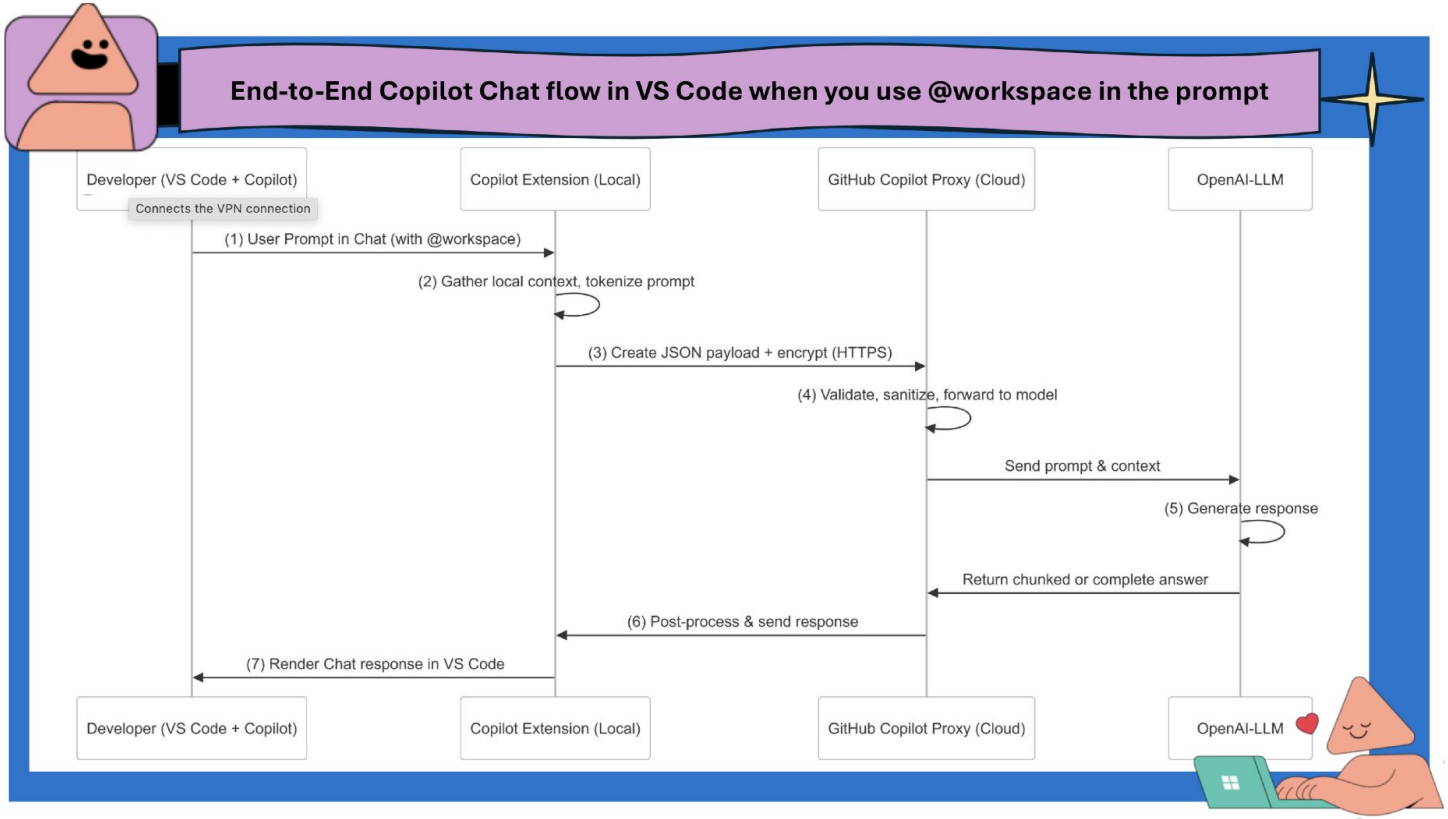
AI CODING TOOLS – COMPARISON TABLE

	Codex	CLAUDE CODE	GitHub Copilot
01 Coding Style	Task Driven	Reasoning Driven	Suggestions Driven
02 Interaction	Low	High	Continuous
03 IDE Integration	Limited	Moderate	High
04 Use For	Automation and Delegation use caes	Complex Reasoning and Algorithms	Day to Day coding
05 Speed	High	Medium	High
06 Learning Aid	Low	High	Medium

Github Copilot - AI Agent Product

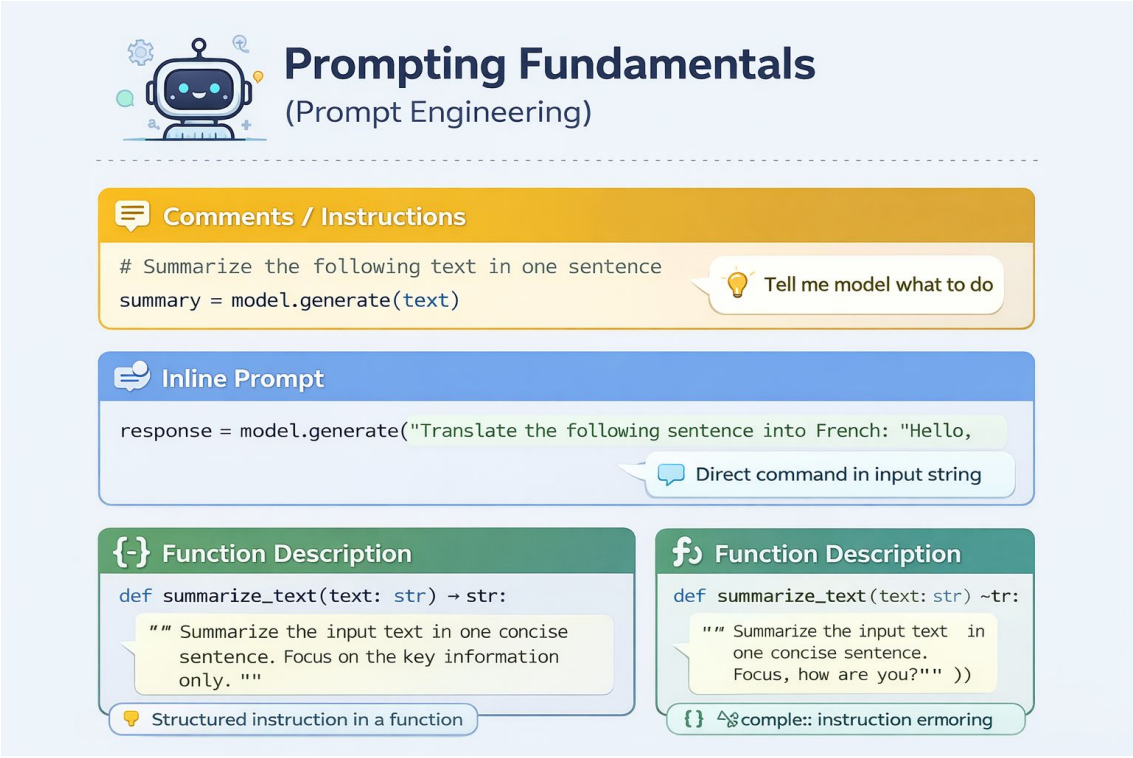


Github Copilot - How it Works?



Github Copilot - Prompt Techniques #1

Since we use general model to do specific tasks, LLM (brains) is stuck to understand it like us Human ^^



Github Copilot - Prompt Techniques #2

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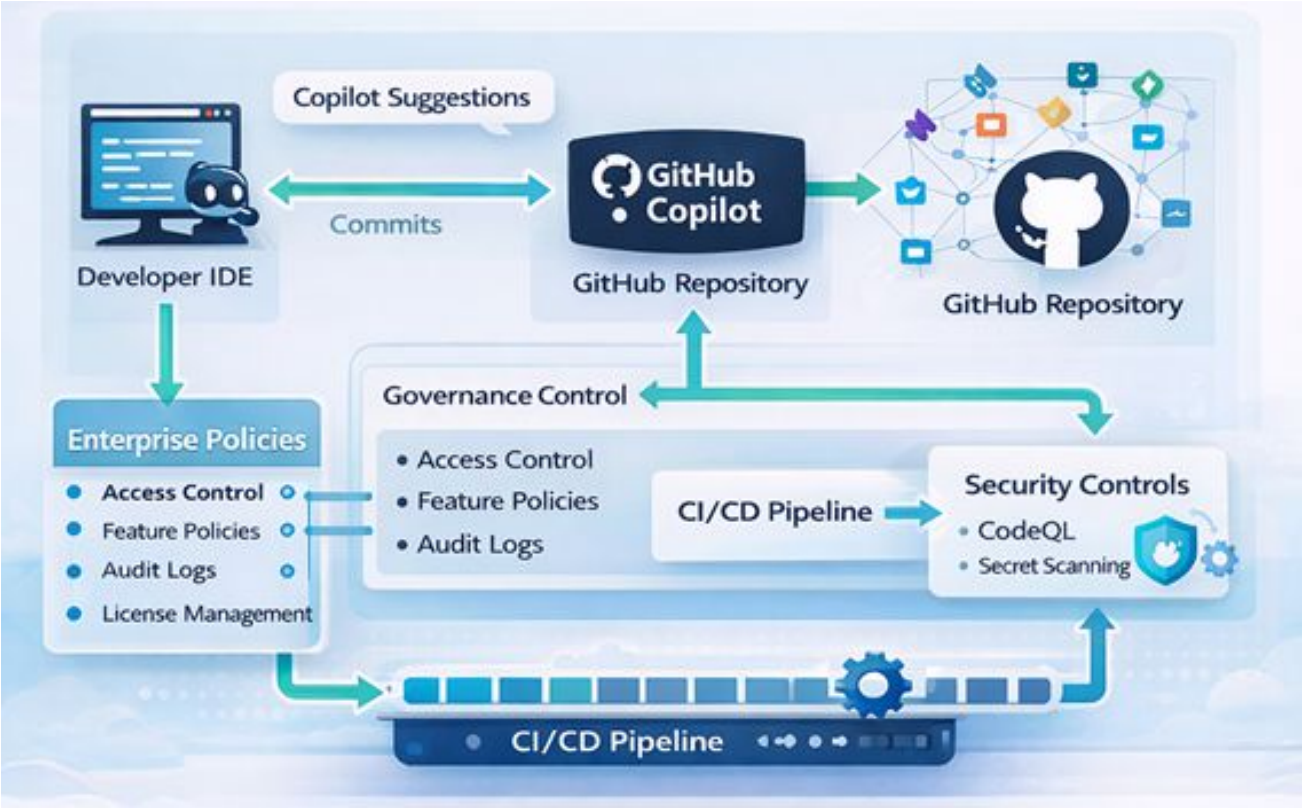
	Zero-shot prompting
Overview	o Refers to a scenario when predictions are produced without any explicit or supplemental examples o Excels at– • Classification tasks like Sentiment analysis or spam detection • Text transformation tasks like translation or summarization, and basic text synthesis • Add text here
Prompt input	"What is the sentiment of the following sentence: 'I had a tiring day at the work'?"
Model's response	"The sentiment of the sentence is negative."

	Few-shot prompting
Overview	o Uses a small number of instances, often between two and five, to direct the model's output o Intended to point the model in the direction of higher performance when dealing with more context-specific issues o Allow the model to – • More effectively tune its replies • Improve the precision of its predictions by providing a snapshot of the intended outcome
Prompt input	Write a rhymed couplet about moonlight. Example 1: 'Under moonlight's gentle hue, Stars twinkle as the night imbues.' Example 2: 'Moonlit night, serene and still, Whispers secrets, nature's thrill.' Now, write a rhymed couplet about sunshine.
Model's response	'Sunshine dances, bright and clear, Chasing shadows, spreading cheer.'

This slide is 100% editable. Adapt it to your needs and capture your audience's attention.

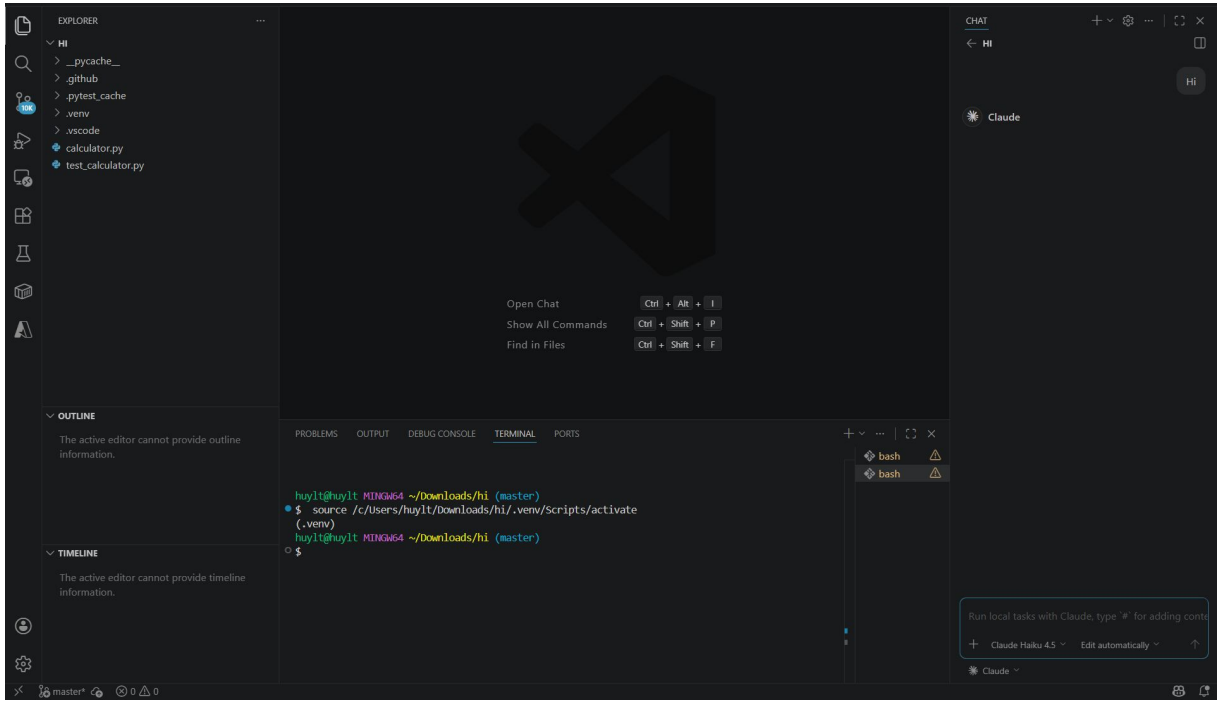
Github Copilot - AI Responsibility and Governance



Github Copilot - YOUR TURN #1

Pls follow guide

1. Install VS Code
2. Install Github Copilot Extension
3. Sign Github into VS Code
4. Make First Conversation Say “Hiiiii”
5. Change Brains to GPT
6. **#Prompt1** ask Agent to build simple function math (plus, minus, divide, multiple) in Java or Python or Node



Github Copilot - YOUR TURN #2

7. **#Prompt2:** Add Few Shoots Prompts “Pls follow style always have comments #VTI in function”

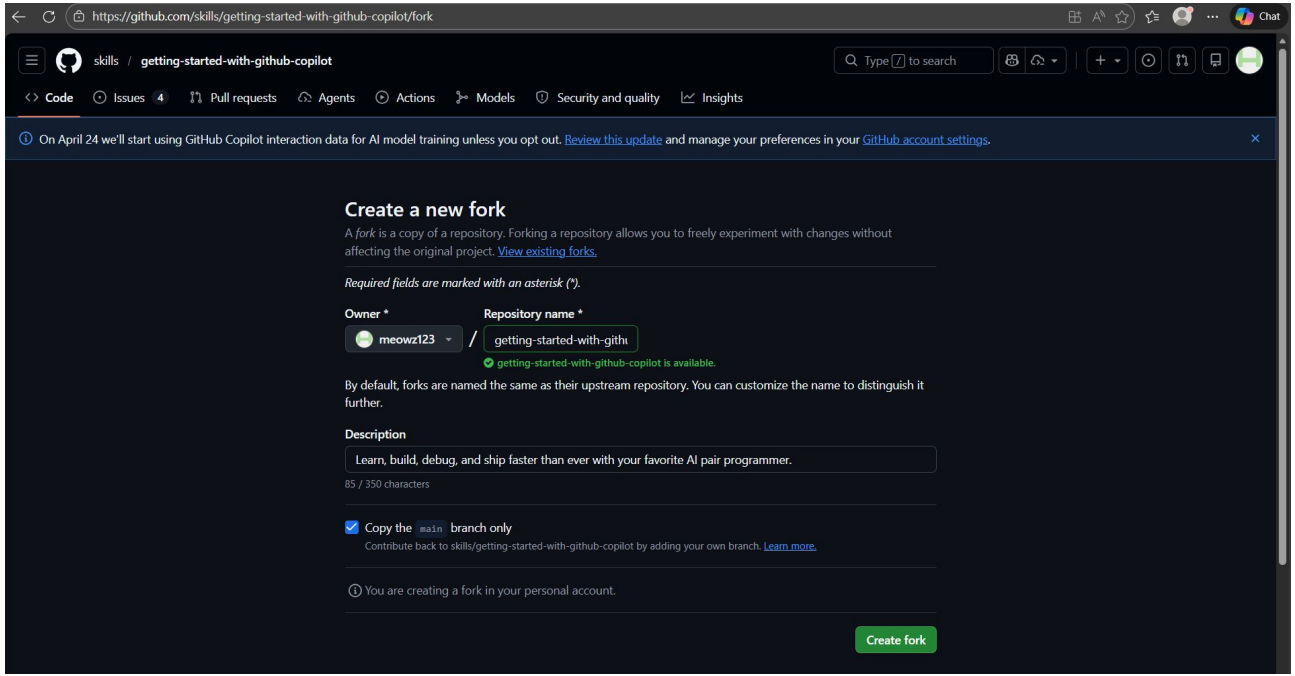
8. Fork it Repo “GitHub - meowz123/session_1_school_manager · GitHub”

9. **#Prompt3:** Ask Github Copilot to pull it into folder

10. #Install Markdown Preview Enhanced

11. Open .github/steps

12. DO These Steps



Github Copilot - Review

Review

Congratulations, you've completed this exercise and learned a lot about GitHub Copilot!

Here's a recap of the GitHub Copilot features you learned:

- **Ask Mode**: Used @workspace to explore and understand your codebase
- **Inline suggestions**: Completed code with Tab acceptance
- **Inline Chat**: Generated code and data with Ctrl/Cmd + I
- **Agent Mode**: Built features autonomously
- **Plan Agent**: Drafted a plan, answered questions, and started implementation
- **GitHub integration**: Generated commit messages, PR summaries, and code reviews

5. Build Application AI

Pets workshop

This repository contains the project for three guided workshops to explore various GitHub features. The project is a website for a fictional dog shelter, with a Flask backend using SQLAlchemy and Astro frontend using Svelte for dynamic pages.

Ready to get started? Let's go! The workshop scenario imagines you as a developer volunteering your time for a pet adoption center. You've been asked to add a filter to the website to allow people to limit their search results by breed and adoption status. You'll work over the next 5 exercises to perform the tasks!

1. Clone the repository and start the app for the workshop.
2. Add an endpoint to the server to list all breeds.
3. Explore the project to get a better understanding of what needs to be done.
4. Create custom instructions to ensure Copilot chat has additional context.
5. Add the new feature to the website, and ensure it works!

Build Application AI - YOUR TURN #4

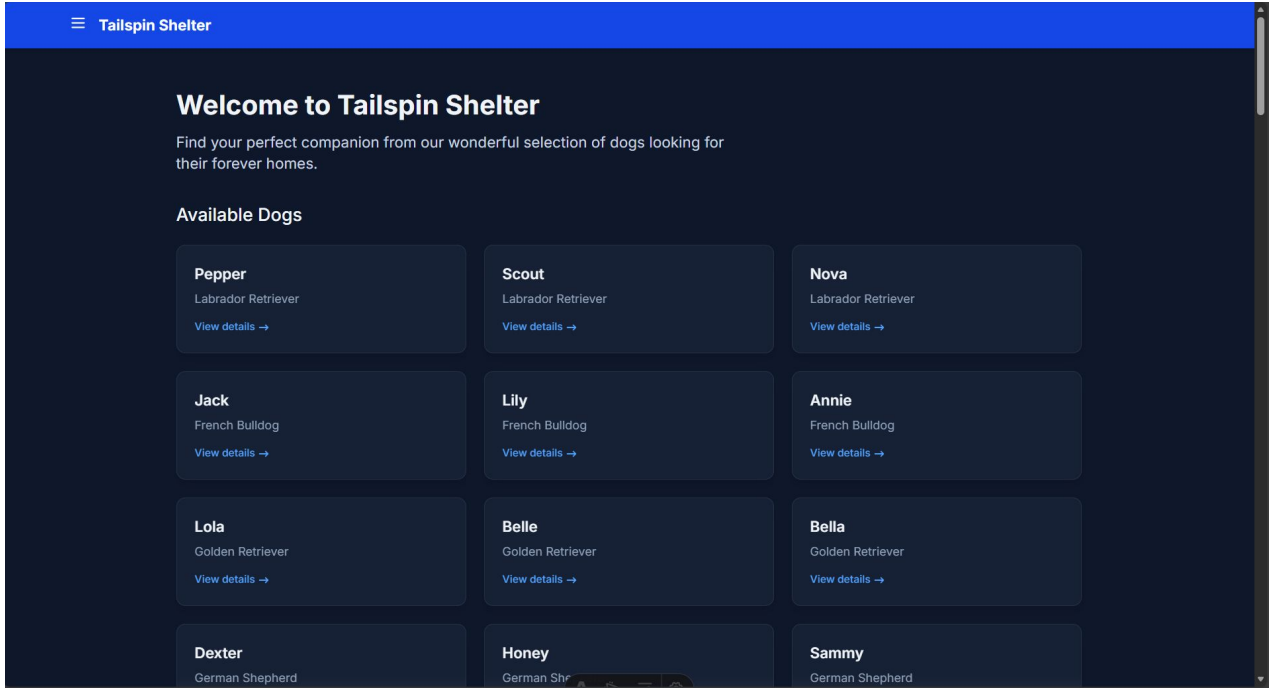
1. Fork it Repo [GitHub - meowz123/session_1_pet_workshop · GitHub]

2. Ask Github Copilot overview projects

3. Open pets-workshop /content/1-hour

4. User requirements want to have search button on right top corner

5. User requirements want to images dogs into UI easy for them to run biz



6. Recap Session 1 - Exercise Next Session

Recap Session 1 - Basic Ideas

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VS Studio Code

🧩 Editor & development environment

💻 Terminal, Debugger, Extensions

🌿 Git integration

Git Hub Copilot

🤖 AI coding assistant

⚡ Code suggestions & auto-complete

🛠️ Helps write & refactor code

Build First Application

🚀 Simple AI chat app

✍️ Use Copilot to generate code

🐞 Run, test, and debug

Exercise Session 1 - I Challenge Jabil

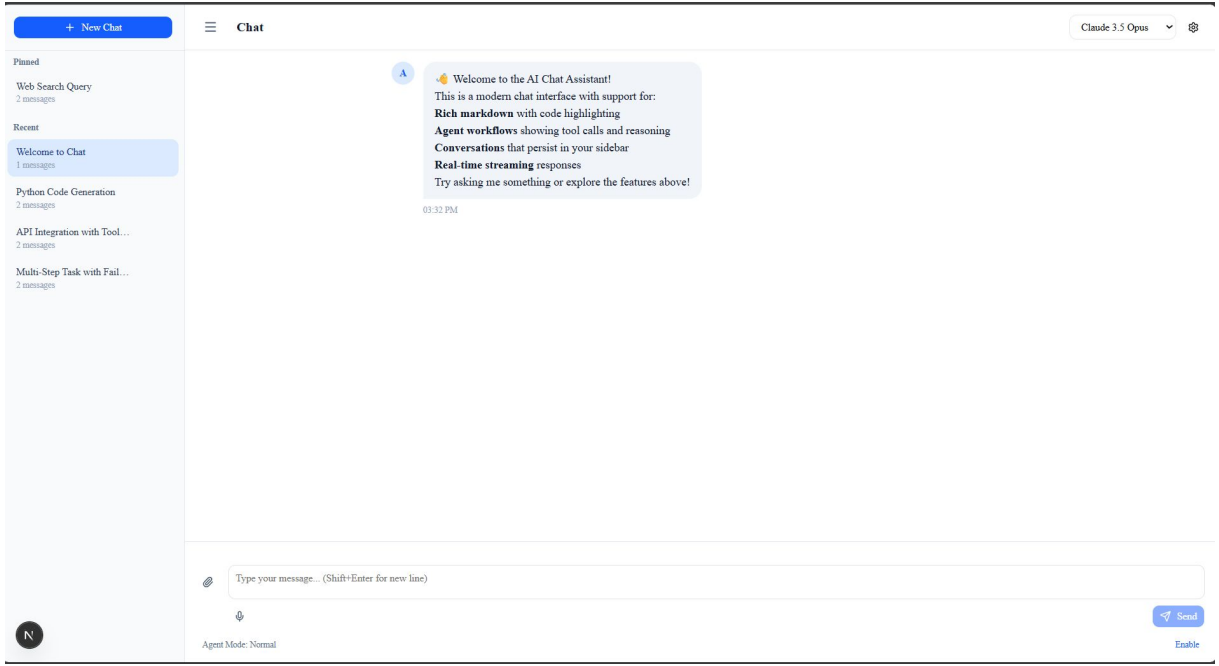
Prompt It !!!

Functional:

- Chat with AI
- Send / receive messages
- View chat history

Advanced:

- Login / logout
- Save chats (DB)
- File upload + Q&A
- Model selection
- UI: dark/light mode



Thank you

References

1. VS Code User Interface Guide
2. GitHub Copilot in VS Code Features
3. Copilot Workspace Context (How it works)
4. Microsoft Learn – GitHub Copilot Training
5. GitHub Copilot Documentation Guide
6. Copilot Code Referencing Explained
7. Copilot Instructions vs Prompts vs Agents